

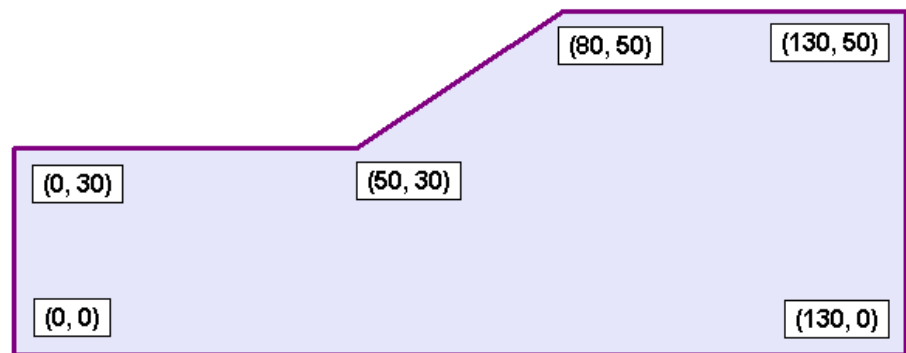
Shear Strength Reduction Analysis

In this tutorial, *Phase2* is used to determine the safety factor of a simple homogeneous slope using the shear strength reduction (SSR) method. This tutorial covers the basics of setting up a model for an SSR analysis in *Phase2*, and interpreting the SSR analysis results.

Topics Covered

- Project Settings
- Shear Strength Reduction
- Boundary Conditions
- Field Stress
- SSR Analysis Results
- Critical SRF

Geometry



Model7

Start the *Phase2* Model program.

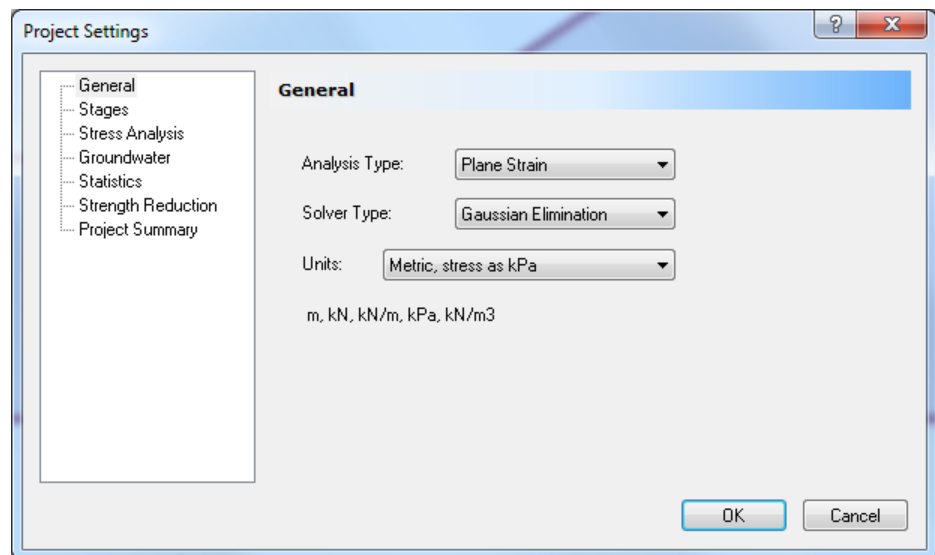
Project Settings



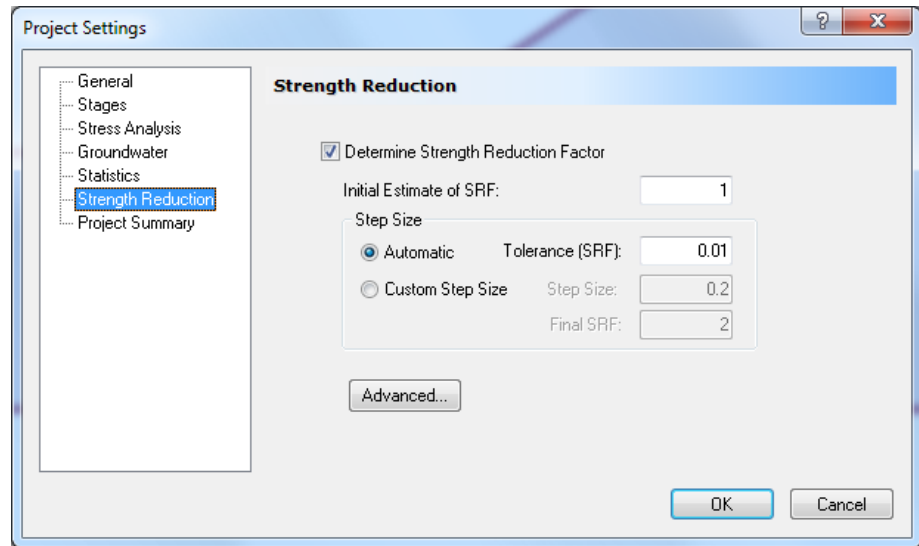
Open the Project Settings dialog from the toolbar or the Analysis menu. Under the General page, define the units as being “Metric, stress as kPa”. Do not change the number of stages and do not exit the dialog.

Note: if you have a multi-stage model, the *Phase2* strength reduction analysis is only carried out at the final stage of the model. If you want to do SSR at an intermediate stage of a multi-stage model, you will have to remove the stages after the stage of interest. You can do this by simply rolling back the number of stages in the Project Settings dialog.

SSR should be used to determine the factor of safety against failure at a particular point in time (i.e. at a particular Stage). Thus, we only do SSR at one stage, not for each stage (of a multi-stage model).



In the Project Settings dialog, select the Strength Reduction page. Turn on the **Determine Strength Reduction Factor** checkbox. This enables the SSR analysis. Leave the various SSR settings at the default values. Close the Project Settings dialog by pressing the OK button.



Boundaries



This model only requires an External boundary to define the geometry. Select the **Add External** option in the Boundaries menu and enter the coordinates shown in the figure at the beginning of this tutorial.

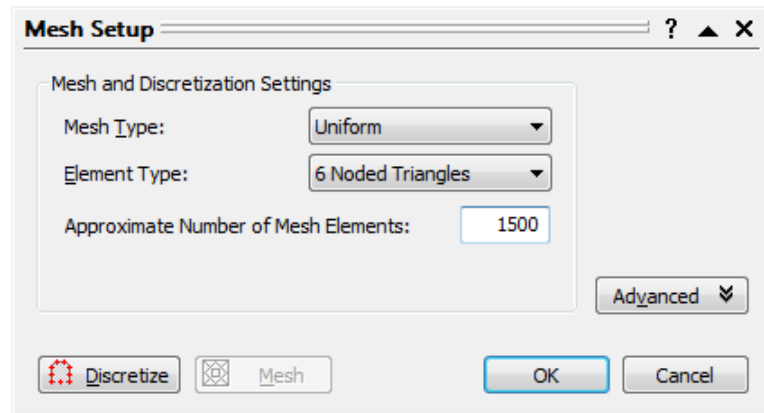
Mesh

Now generate the finite element mesh. Before we do this, let's define the parameters (type of mesh, number of elements, type of element) used in the meshing process.

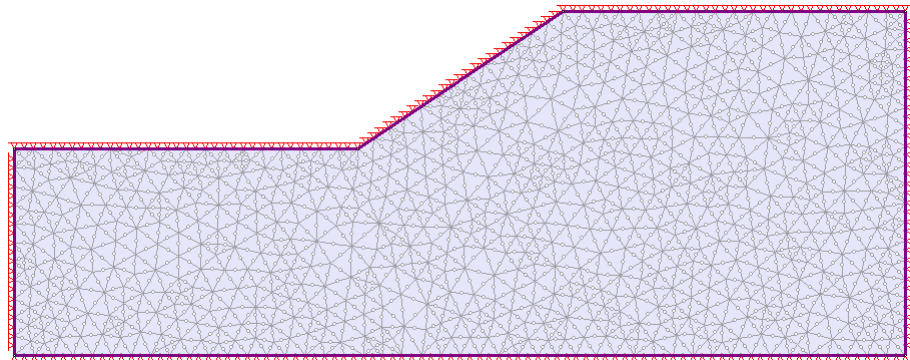


1. Select the **Mesh Setup** option in the **Mesh** menu.
2. In the Mesh Setup dialog, change the Mesh Type to Uniform, the Element Type to 6 Noded Triangles and the number of elements to 1500.
3. Close the Mesh Setup dialog by selecting the OK button.

Based on our experience with numerous SSR models, we suggest using a uniform mesh with 6 noded triangles for all SSR analyses. The number of elements depends on the complexity of your model. If it is a simple model, 1500 elements should be adequate. If the model is more complicated, then a larger number of elements may be necessary (e.g. 3000). You can always try different mesh densities to make sure you are using enough elements to capture the correct behavior.



Mesh the slope by selecting the **Discretize and Mesh** option from the toolbar or the **Mesh** menu.



Mesh and default boundary conditions

Boundary Conditions

Now we can set the boundary conditions. The portion of the external boundary representing the ground surface (0,30 to 50,30 to 80,50 to 130,50) must be free to move in any direction.



1. Select the **Free** option in the **Displacements** menu.
2. Use the mouse to select the three line segments defining the ground surface of the slope.
3. Right-click and select Done Selection.

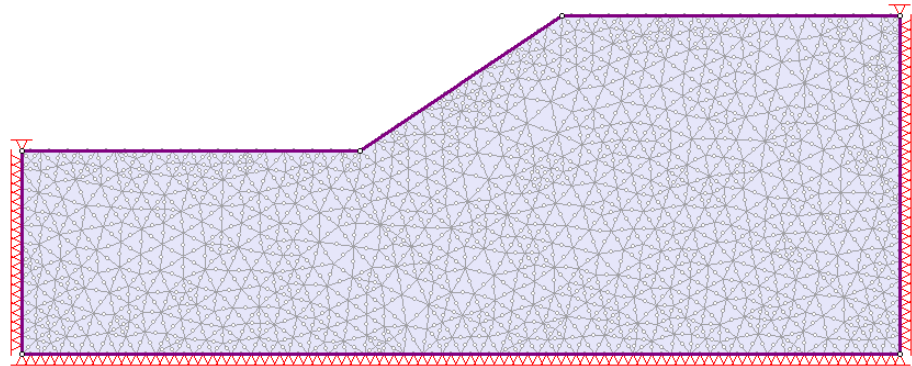
TIP: you can also right-click on a boundary to define its boundary conditions.

The slope surface is now free, however, this process has also freed the vertices at the upper left and upper right corners of the model. Since these edges should be restrained, we have to make sure that these two corners are restrained.

Let's use the right-click shortcut to assign boundary conditions:

1. Right-click the mouse directly on the vertex at (0,30). From the popup menu select the Restrain X,Y option.
2. Right-click the mouse directly on the vertex at (130,50). From the popup menu select the Restrain X,Y option.

The displacement boundary conditions are now correctly applied.



Free boundary condition applied to ground surface

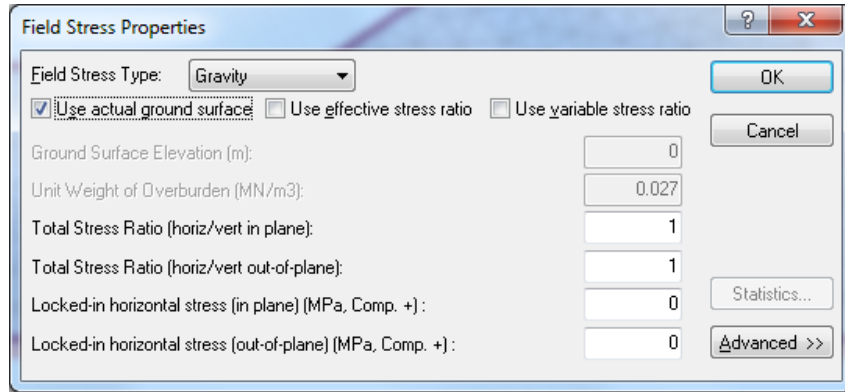
NOTE: in general, the displacement boundary conditions for an SSR analysis of a slope will be a Free ground surface, and Fixed XY for the remainder of the external boundary.

Field Stress

Now define the in-situ stress field.



1. Select the **Field Stress** option in the **Loading** menu.
2. Change the Field Stress Type from Constant to Gravity (gravitational stress distribution throughout the slope).
3. Check the **Use actual ground surface** checkbox. By using this option, the program will automatically determine the ground surface above every finite-element and define the vertical stress in the element based on the weight of material above it.
4. Leave the horizontal stress ratios as 1, meaning hydrostatic initial stresses (i.e. horizontal stress = vertical stress). If you know the actual horizontal stress ratio when doing your own slope model, you can use this information. However, the horizontal stress distribution within a slope is rarely known, so leaving the default hydrostatic stress field has shown to be a good assumption.



The 'Field Stress Properties' dialog box is shown. It has a title bar with a question mark and a close button. The 'Field Stress Type' is set to 'Gravity'. There are three checkboxes: 'Use actual ground surface' (checked), 'Use effective stress ratio' (unchecked), and 'Use variable stress ratio' (unchecked). Below these are input fields for 'Ground Surface Elevation (m)' (0), 'Unit Weight of Overburden (MN/m3)' (0.027), 'Total Stress Ratio (horiz/vert in plane)' (1), 'Total Stress Ratio (horiz/vert out-of-plane)' (1), 'Locked-in horizontal stress (in plane) (MPa, Comp. +)' (0), and 'Locked-in horizontal stress (out-of-plane) (MPa, Comp. +)' (0). On the right side, there are buttons for 'OK', 'Cancel', 'Statistics...', and 'Advanced >>'.

Material Properties

Define the material properties of the soil that comprises the slope.



Select **Define Materials** from the toolbar or the Properties menu.

Type Till for the name. Make sure the Initial Element Loading is set to Field Stress & Body Force (both in-situ stress and material self weight are applied). Enter 19 kN/m³ for the Unit Weight. For Elastic Properties, enter 50000 kPa for the Young's Modulus and 0.4 for the Poisson ratio. For Strength Parameters, make sure the Failure Criterion is set to Mohr-Coulomb. Set the Material Type to Plastic, meaning the material can yield/fail. Set the peak and residual Tensile Strength to 5 kPa. Set the peak and residual Cohesion to 5 kPa. Set the peak and residual Friction Angle to 30°. Leave the dilation angle at 0° (no volume increase when sheared, non-associated flow rule). Press the OK button to save the properties and close the dialog.

The screenshot shows the 'Define Material Properties' dialog box for a material named 'Till'. The left pane lists materials from Till to Material 29. The right pane shows the following settings:

- Name:** Till
- Material Color:** (light blue)
- Initial Element Loading:** Field Stress & Body Force
- Unit Weight:** (kN/m³): 19
- Elastic Properties:**
 - Elastic Type:** Isotropic
 - Poisson's Ratio:** 0.4
 - Young's Modulus (kPa):** 50000
 - Young's Modulus (resid) (kPa):** 20000
 - E1 (kPa):** 20000
 - E2 (kPa):** 20000
 - Ez (kPa):** 20000
 - v12:** 0.2
 - v13:** 0.2
 - v23:** 0.2
- Strength Parameters:**
 - Failure Criterion:** Mohr Coulomb
 - Material Type:** Plastic
 - Tensile Strength (peak) (kPa):** 5
 - Dilation Angle (deg):** 0
 - Fric. Angle (peak) (deg):** 30
 - Fric. Angle (resid) (deg):** 30
 - Cohesion (peak) (kPa):** 5
 - Cohesion (resid) (kPa):** 5
 - Tensile Strength (resid) (kPa):** 5
- Stage Properties:** (unchecked)
- Datum Dependent:** (unchecked)
- Unsaturation Shear Strength:**
 - Phi b:** 0
 - Air Entry (kPa):** 0

Buttons at the bottom include 'Copy To...', 'Statistics...', 'Show only properties used in model', 'Apply SSR' (checked), 'OK', and 'Cancel'.

You have completed the definition of the model. Save the model using the Save option in the File menu.

Compute



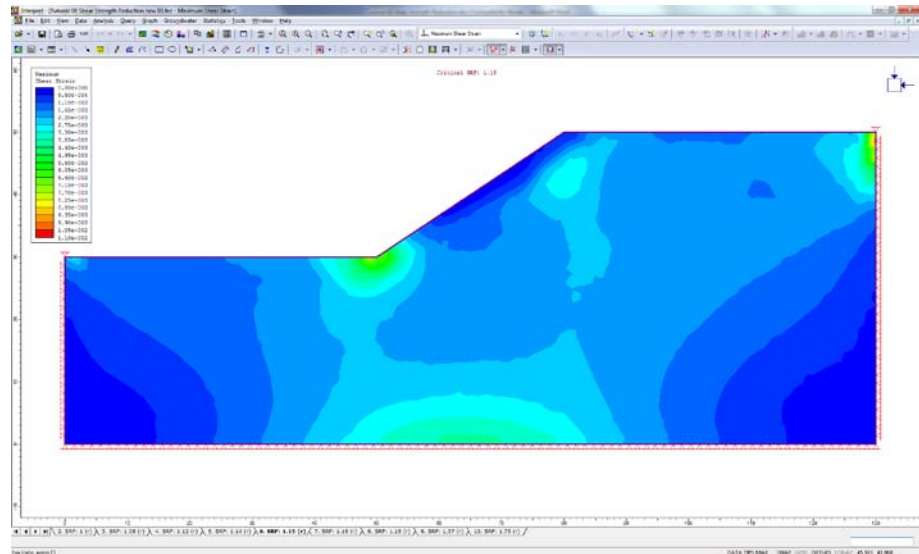
Run the model using the Compute option in the Analysis menu. The analysis should take under a few minutes to run.

Once the model has finished computing (Compute dialog closes), select the Interpret option in the Analysis menu to view the results.

Interpret

After you select the Interpret option, the Interpret program starts and reads the results of the analysis. You should see the following screen with the critical strength reduction factor (SRF) of 1.15 displayed at the top of the window.

NOTE: if this same model is computed in *Slide* the limit equilibrium safety factor is 1.14 compared to a critical SRF of 1.15 in *Phase2*.

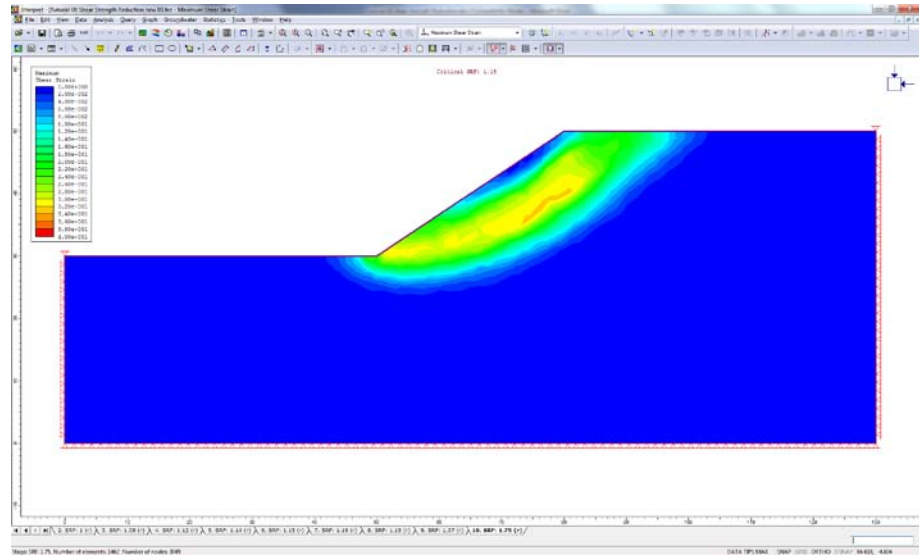


Note the different values of SRF (strength reduction factor) in the tabs along the bottom of the screen. The tab that is selected by default is the critical SRF. By default the maximum shear strain dataset is selected and contoured. Maximum shear strain will give you a good indication of where slip is occurring, especially if you change the view to higher SRF values.

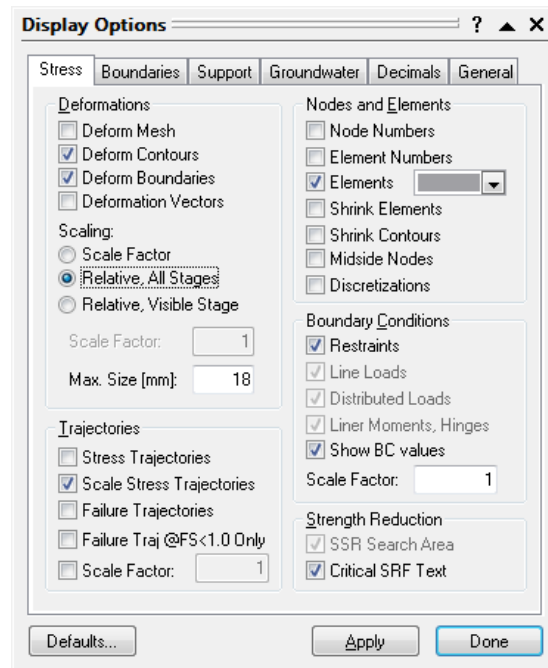
By cycling through the various SRF tabs, you get a good indication of the progression of failure through your slope as the shear strength is reduced.

TIP: you can also change the viewing stage by selecting the Page Up / Page Down keys, or by placing the mouse cursor over the stage tabs and rotating the mouse wheel.

Change the SRF to 1.75 by clicking on the SRF: 1.75 tab. Note the well formed shear band. The view should look like the following image.



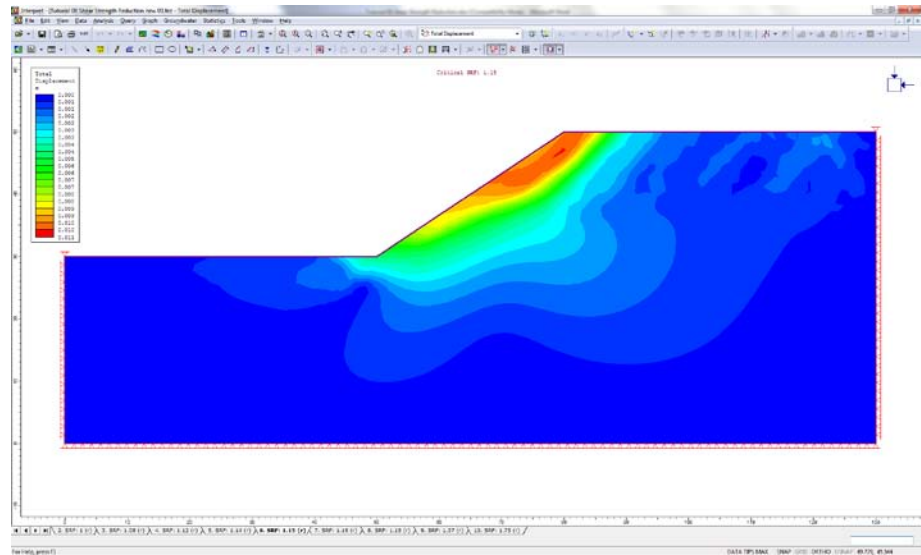
Now switch to SRF: 1. Another useful feature is the ability to animate the progression of failure. Select Display Options in the View menu. Choose the Stress tab. Turn on the Deform Contours and Deform Boundaries options. Select the **Relative, All Stages** scaling option. Select Done to save and exit the dialog. Now choose the Animate Tabs option in the Data menu.



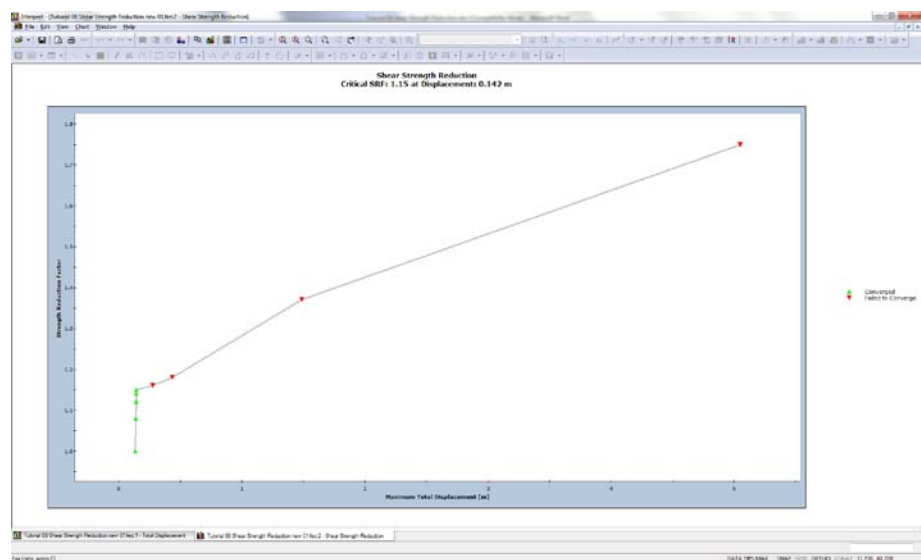
Press the Escape key to stop the animation.

NOTE: the timing of the Animation can be specified in the Stage Settings dialog in the Data menu.

Turn off deformed contours and boundaries and select the critical SRF tab (SRF: 1.15). Another dataset of interest is Total Displacement. Use the combobox in the toolbar to select the Total Displacement dataset. The displacement contours clearly highlight the zone of failure.



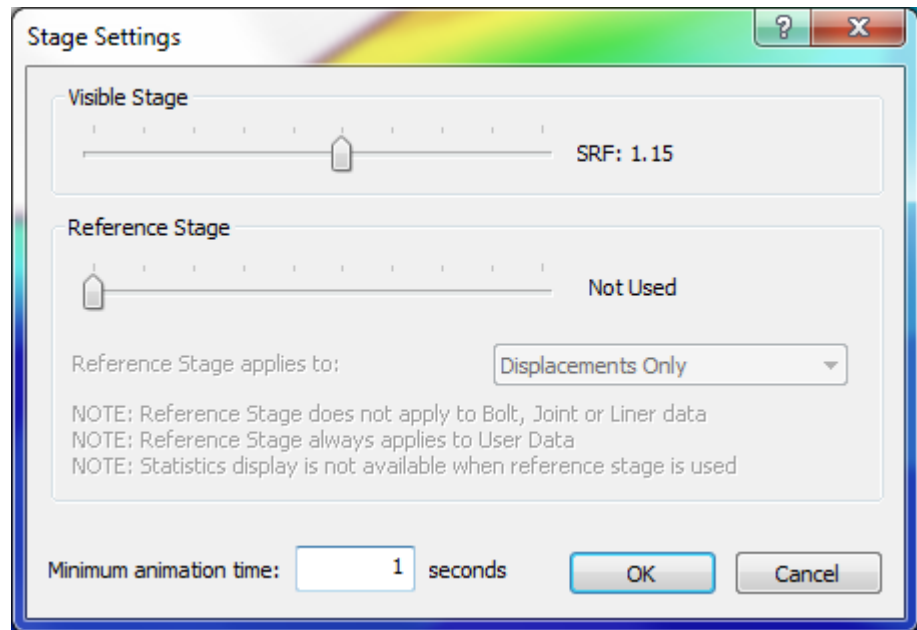
Another important feature when doing an SSR analysis is the ability to plot maximum deformation versus SRF. As the SRF is increased, the strength properties are decreased. As the strength decreases the maximum displacement increases. At some point, the slope will fail, and deformations will increase rapidly and the finite-element analysis will not converge. It is this point of non-convergence that defines the critical SRF. To view this plot, select the **Graph Shear Strength Reduction** option in the **Graph** menu. The following plot is generated. Notice the inflection in the displacements and the point where the solution does not converge.



Close the graph.

You may have noticed that the stress analysis results of Stage 1 are not available to look at. Only the data for the different SRF values are available. It is possible to view the results from the stage or stages prior to the SSR analysis. To view the results from these stages:

1. Select the Stage Settings option in the Data menu.
2. Move the reference stage slider all the way to the left so that it reads “Not Used”.
3. Select OK.



A Stage 1 tab now exists along with the SRF tabs at the bottom of the view. You can now see the results of Stage 1 by selecting the Stage 1 tab.

You may wonder why SRF: 1 was used as a reference stage for displacements. In order to factor out the elastic displacements due to rebound and stress redistribution (initial stresses and body forces are rarely in equilibrium to start with), the displacements at the minimum SRF stage are factored out. Thus all SRF displacements are relative to the displacements that occur at the minimum SRF stage.

Note: the results of Stage 1 and SRF: 1 are slightly different even though the material properties are the same for both. The reason is the different tolerance and iteration count used for each. To accelerate the SSR analysis, speed optimized values of tolerance and number of iterations are used.

- You can edit the tolerance and number of iterations used in the SSR analysis in the advanced options of the Strength Reduction page in the Project Settings dialog of the *Phase2* modeler.
- You can edit the tolerance and number of iterations used for stages prior to the SSR analysis (in this case Stage 1), under the Stress Analysis page in Project Settings.

This concludes the tutorial, you may now exit the *Phase2* Interpret and *Phase2* Model programs.